gistration Number: RA2211026030045

SF

SRM Institute of Science and Technology

Department of Computer Science and Engineering

Delhi – Meerut Road, Sikri Kalan, Ghaziabad, Uttar Pradesh – 201204

Academic Year: 2025-26 (ODD)

: **Internal Examination I**
se Code & Title: **21CSC401J & Deep Learning Techniques**
& Sem : **IV & VII**

Date & Session: 03-09-2025
Duration: 1 Hour 30 Minut
Max. Marks: 50

Part - A

ver all questions **(10 x 1 = 10 Ma**

Question	Marks	BL	C
The McCulloch-Pitts neuron is primarily used to model: a) Biological action potentials b) Logical functions c) Gradient descent updates d) Activation functions	1	1	1
The primary purpose of a McCulloch-Pitts neuron? a) Generate action potentials b) Perform logical operations c) Apply gradient descent d) Serve as an activation function	1	1	
Why a single perceptron fails to learn XOR. a) Lack of convergence b) Non-linear separability c) Wrong loss function d) High variance	1	1	
Sigmoid neuron output for input=0 is: a) 0 b) 0.25 c) 0.5 d) 1	1	2	
The perceptron output for AND gate with input (1,0) is. a) 0 b) 1 c) Undefined d) Depends on learning rate	1	2	
The vanishing gradient problem is most associated with: a) ReLU b) Sigmoid c) Tanh d) Leaky ReLU	1	1	
RMSProp improves AdaGrad by: a) Lower cost b) Handling sparse features c) Avoiding aggressive decay d) Only convex use	1	1	
The main purpose of early stopping in training neural networks is: a) Prevent overfitting b) Reduce gradient explosion	1	2	

	c) Increase dataset size d) Improve activation function				
9	The main advantage of using mini-batch gradient descent compared to batch and stochastic gradient descent a) Provides a good balance between memory usage and speed of convergence b) Always reaches the global minimum c) Does not require any parameter tuning d) Completely removes randomness in updates	1	1	2	[illegible]
10	Adam optimizer combines the features of a) Momentum & AdaGrad b) RMSProp & Nesterov c) Gradient descent & Mini-batch d) Mini-batch & Momentum	1	1	2	2

Part B

Answer any two questions — **4M x 2Q = 8 Marks**

11	Illustrate the role of bias in a perceptron. Why is it important in learning decision boundaries.	4	3	1	2
12	Outline the backpropagation algorithm for training a neural network.	4	4	1	2
13	Demonstrate the importance of activation functions in neural networks. Why can't we train deep models without them?	4	3	1	2

Part C

Answer any two questions — **4M x 2Q = 8 Marks**

14	Determine the concept of vanishing and exploding gradients in deep neural networks.	4	3	2	2
15	Illustrate mini-batch gradient descent, batch gradient descent, and stochastic gradient descent.	4	3	2	2
16	Classify the concept of regularization in deep learning.	4	4	2	[illegible]

Part D

Answer all questions — **12M x 2Q = 24 Mar**

17	(A) Describe the gradient descent algorithms. Compare advantages, and limitations with examples	12	4	1
	(OR)			
	(B) Classify the Backpropagation algorithm in detail. explain how the error is propagated backwards layer by layer to update the weights. Support your answer with a simple two-layer example	12	4	1
18	(A) Discuss the impact of Hyperparameters on neural network training. Explain learning rate, batch size, momentum, epochs, and weight initialization. How do they affect training accuracy and convergence?	12	4	2
	(OR)			
	(B) Explain Overfitting in deep networks. Why does it occur, and how can we reduce it using data augmentation, dropout, regularization, and cross-validation?	12	4	2

Registration Number: RA2211026030045

SRM

SRM Institute of Science and Technology

Department of Computer Science and Engineering

Delhi - Meerut Road, Sikri Kalan, Ghaziabad, Uttar Pradesh - 201204

Academic Year: 2025-26 (ODD)

Test : **Internal Examination I**
Course Code & Title : 21CSE451T & Pattern Recognition Techniques
Year & Sem : **IV/ VII**

Date & Session : 01.09.2025 AN
Duration: 1 Hour 30 Minutes
Max. Marks: 50

Part - A

Answer all questions **(10Q x 1M = 10 Marks)**

Q. No	Question	Marks	BL	CO	PO
1	Conditional probability P(A|B) is defined as: a) P(A)/P(B) b) P(B)/P(A) c) P(A∩B)/P(B) d) P(A)+P(B)	1	L3	1	1
2	In supervised pattern recognition, the training data must contain: a) Only input features b) Only output labels c) Both input features and corresponding labels d) No labels at all	1	L2	1	1
3	In pattern recognition, assuming a normal (Gaussian) density is helpful because it: a) Simplifies the estimation of probability density functions b) Always guarantees correct classification c) Eliminates the requirement for training data d) Ensures prior probabilities are equal	1	L3	1	1
4	In discriminant functions $g_i(x)$, classification is done by: a) Choosing the smallest value of $g_i(x)$ b) Choosing the largest value of $g_i(x)$ c) Randomly choosing a class d) Choosing based on prior only	1	L3	1	1
5	In pattern recognition, classification refers to: a) Dividing data into meaningful regions b) Assigning input data to one of several predefined categories c) Extracting key features from raw data d) Reducing dimensionality of data	1	L1	1	1
6	In parameter estimation, the maximum likelihood method finds parameters that: a) Maximize the posterior probability b) Minimize the loss function c) Maximize the likelihood of the observed data d) Minimize the risk	1	L4	2	2
7	The function $p(x\|\omega_i)$ is also known as: a) Likelihood function b) Posterior function c) Prior function d) Decision function	1	L3	2	2
8	PCA finds directions in the data that: a) Maximize the variance b) Minimize the variance c) Minimize reconstruction error only d) Randomly distribute the data	1	L2	2	1

9	A Hidden Markov Model consists of: a) Only observed states b) Hidden states and observed outputs c) Only hidden states d) Deterministic transitions	1	L2	2	1
10	Multiple Discriminant Analysis (MDA) is used for: a) Reducing dimensionality and maximizing class separability for two classes only b) Reducing dimensionality and maximizing class separability for multiple classes c) Estimating probability densities d) Clustering unlabeled data	1	L4	2	2

Part B

Answer any two questions **4M x 2Q = 8 Marks**

11	Summarize future trends in the use of supervised, unsupervised, and reinforcement learning for pattern recognition.	4	L2	1	1
12	Derive how the discriminant functions used as classifier in univariate case?	4	L3	1	1
13	State the fundamental rules of probability theory.	4	L4	1	1

Part C

Answer any two questions **4M x 2Q = 8 Marks**

14	How to deal with the effect of high-dimensionality features to complexity?	4	L4	2	1
15	Elaborate two approaches for parameter estimation method.	4	L3	2	2
16	Why is FLD considered a supervised dimensionality reduction method?	4	L4	2	2

Part D

Answer all questions **12M x 2Q = 24 Marks**

17	(A) Suppose you are designing a medical image classification system. Describe how each component would be implemented.	12	L3	1	1
	(OR)				
	(B) i) In a group of 100 sports car buyers, 40 bought alarm systems, 30 purchased bucket seats, and 20 purchased an alarm system and bucket seats. Find the probability that a buyer who purchased an alarm system also purchased bucket seats. ii) A survey was conducted in a class of 30 students. 20 students preferred chocolate ice cream, and 10 preferred vanilla. What is the empirical probability that a randomly selected student would prefer chocolate?	6+6	L4	1	1
18	(A) Derive the algorithm used for evaluating the likelihood of an observation sequence in HMMs. State the problems associated with HMM model.	12	L4	2	2
	(OR)				
	(B) How the E-step and M-step mathematically be iteratively applied in pattern recognition approaches?	12	L3	2	2

Registration Number:	R	A	2	2	1	1	0	2	6	0	3	0	0	4	5

SRM
DELHI-NCR CAMPUS, GHAZIABAD (U.P)

SRM Institute of Science and Technology

Department of Computer Science and Engineering

Delhi – Meerut Road, Sikri Kalan, Ghaziabad, Uttar Pradesh – 201204

Academic Year: 2025-26 (ODD)

Test : **Internal Examination I**
Course Code & Title : **21MEO112T/Renewable Energy Sources and Application**
Year & Sem : **IV/VII**

Date & Session: 08/09/25 / AN
Duration: 1 Hour 30 Minutes
Max. Marks: 50

Part - A

Answer all questions **(10 x 1 = 10 Marks)**

Q. No	Question	Marks	BL	CO	PO
1	**Find the name of device which converts solar energy directly into electrical energy.** a) Solar collector b) Solar heater c) Photovoltaic cell d) Solar concentrator	1	1	1	1
2	**The main advantage of solar energy is** a) High operating cost b) Non-renewable in nature c) Eco-friendly and abundant d) Requires fossil fuels	1	1	1	1
3	**Which type of solar radiation is most responsible for electricity generation in solar cells?** a) Infrared b) Ultraviolet c) Visible light d) X-rays	1	1	1	1
4	**The unit of solar radiation intensity is:** a) Watt per square meter (W/m^2) b) Joule per second c) Lux d) Candela	1	1	1	1
5	**The approximate average solar constant at the Earth's atmosphere is** a) 100 W/m^2 b) 340 W/m^2 c) 1367 W/m^2 d) 2000 W/m^2	1	1	1	1
6	**Wind energy is a form of:** a) Kinetic energy b) Potential energy c) Thermal energy d) Nuclear energy	1	1	2	7
7	**The largest producer of wind energy country (as of 2023) is** a) USA b) China c) India d) Germany	1	2	2	7

8	**Following is used to measure wind speed** a) Anemometer b) Barometer c) Hygrometer d) Altimeter	1	2	2
9	**The main environmental concern of wind farms is:** a) Carbon emissions b) Noise and impact on birds c) Radioactive waste d) Water pollution	1	2	2
10	**The power available in the wind is proportional to:** a) Wind speed (v) b) v^2 c) v^3 d) v^4	1	1	2

Part B

Answer any two questions — **4M x 2Q = 8 Marks**

11	Explain the attenuation of solar radiation.	4	2	1
12	Describe the working principles of a solar pond and outline its various applications.	4	2	1
13	Explain the measurement device of solar radiation.	4	2	1

Part C

Answer any two questions — **4M x 2Q = 8 Marks**

14	Describe the term wind energy and origin of wind.	4	2	2
15	Explain the beam, diffuse and global radiation.	4	2	2
16	Describe the nature of wind and its application.	4	2	2

Part D

Answer all questions — **12M x 2Q = 24 Marks**

17	(A) Illustrate the classification of solar concentrators with neat diagrams.	12	3	1
	(OR)			
	(B) Describe the working principle of solar photovoltaic system, components and its application.	12	3	1
18	(A) Illustrate the wind data measurement system.	12	3	2
	(OR)			
	(B) Describe the wind energy scenario in India and the world.	12	3	2

Registration Number: RA2211026030111

SRM Institute of Science and Technology

Department of Computer Science and Engineering

Delhi – Meerut Road, Sikri Kalan, Ghaziabad, Uttar Pradesh – 201204

Academic Year: 2025-26 (ODD)

Test : Internal Examination I		**Date & Session : 02/09/25&AN**
Course Code & Title : 21CSE415T& Fuzzy logic and its applications		**Duration: 1 Hour 30 Minutes**
Year & Sem : 4th &7th		**Max. Marks: 50**

Part - A

Answer all questions **(10 x 1 = 10 Marks)**

Q. No	Question	Marks	BL	CO	PO
1	The composition of two fuzzy relations is performed using a) Intersection and union b) Min-max or max-min operations c) Addition and subtraction d) Differentiation	1	2	1	2
2	Which operation in fuzzy sets corresponds to the intersection of classical sets? a) Maximum b) Minimum c) Addition d) Subtraction	1	1	1	1
3	Specify the role that membership functions do play in fuzzification? a) They convert fuzzy output to crisp output b) They assign degrees of membership to crisp inputs c) They generate fuzzy rules d) They perform defuzzification	1	1	1	1
4	The fuzzy equivalence relation is also known as a) Fuzzy congruence b) Fuzzy similarity relation c) Fuzzy implication d) Fuzzy complement	1	1	1	1
5	The height of a fuzzy set is defined as: a) The total number of elements in the set b) The average membership value c) The maximum membership value among all elements d) The difference between the highest and lowest membership values	1	2	1	2
6	Which matrix in the RLS algorithm helps in weighting the importance of new versus old data? a) Covariance matrix b) Identity matrix c) Hessian matrix d) Transition matrix	1	3	2	3
	In decision-making, rank ordering helps to a) Randomly shuffle options b) Eliminate the need for criteria c) Prioritize alternatives based on preferences or performance d) Guarantee optimal results	1	2	2	2

8 Fuzzy logic is best suited for which type of problem? 1 1 2

a) Precise mathematical equations
b) Problems with binary outcomes only
c) Problems involving uncertainty or linguistic variables
d) High-speed arithmetic operations

9 What does the height of a membership function at a specific point represent? 1 2 2

a) The probability of the input value
b) The degree of membership of that value in the fuzzy set
c) The crisp output value
d) The error rate

10 Identify the potential issue with setting the learning rate too high in gradient descent? 1 3 2

a) The system becomes more accurate
b) The algorithm converges faster
c) The algorithm may overshoot and fail to converge
d) It leads to underfitting

Part B

4M x 2Q = 8 M

Answer any two questions

11 How is the complement of a fuzzy set defined? What is its effect on the membership values? 4 1

12 Compare and contrast between fuzzy tolerance and equivalence relation. 4 2

13 Discuss the concept of support and core of a fuzzy set. How do they relate to the uncertainty and definiteness of the fuzzy set? 4 3

Part C

4M x 2Q = 8

Answer any two questions

14 Give an example of a scenario where modified learning from examples is applied. 4 3

15 How would fuzzy logic handle the linguistic term "moderately warm" in a temperature control system? 4 1

16 Which fuzzy system parameters can be adjusted using the gradient method? 4 2

Part D

12M x 2Q =

Answer all questions

17 (A) Explain the composition of fuzzy relations. Describe different types of composition methods with suitable examples. 12 2

(OR)

(B) Describe different types of membership functions used in fuzzification. How do they affect system performance? 12 1

18 (A) Explain the Batch Least Squares (BLS) algorithm in detail. Discuss its mathematical formulation, working mechanism, advantages, disadvantages, and applications. 12

(OR)

(B) Define inference and explain its role in logical reasoning. What are the types of inference and how do they contribute to knowledge and decision-making? 12